Eesha Chavan **D20A** **15**

Blockchain Experiment 11

AIM: To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

PROCEDURE

Step 1: Update the system

Go to https://shell.cloud.google.com and run the following commands:

```
sudo apt update
sudo apt upgrade
-y
```

```
g2022_anuprita_mhapankar@cloudshell:~$ sudo apt update
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Hit:2 https://download.docker.com/linux/ubuntu noble InRelease
Hit:3 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease
Hit:4 https://packages.cloud.google.com/apt gcsfuse-noble InRelease
Hit:5 https://packages.cloud.google.com/apt cloud-sdk InRelease
Hit:6 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease
Hit:7 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:8 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:9 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:10 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease
Hit:11 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 3,917 B in 6s (639 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
g2022_anuprita_mhapankar@cloudshell:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  libpam-systemd libsystemd-shared libsystemd0 libudev1 qemu-user qemu-user-static systemd systemd-dev systemd-sysv udev
The following packages will be upgraded:
  google-cloud-cli-app-engine-go google-cloud-cli-app-engine-java google-cloud-cli-app-engine-python google-cloud-cli-app-engine-python-extras
  google-cloud-cli-bigtable-emulator google-cloud-cli-cbt google-cloud-cli-cloud-run-proxy google-cloud-cli-datastore-emulator google-cloud-cli-gke-gcloud-auth-plugin
  google-cloud-cli-istioctl google-cloud-cli-kpt google-cloud-cli-local-extract google-cloud-cli-minikube google-cloud-cli-nomos google-cloud-cli-package-go-module
  google-cloud-cli-pubsub-emulator google-cloud-cli-skaffold google-cloud-cli-spanner-cli kubectl libarchive13t64 libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-common libpq-dev
  libpq5 libruby3.2 linux-libc-dev php8.3-bcmath php8.3-cgi php8.3-cli php8.3-common php8.3-dev php8.3-mbstring php8.3-mysql php8.3-opcache php8.3-readline php8.3-xml
  ruby3.2 ruby3.2-dev
38 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
10 not fully installed or removed.
Need to get 603 MB of archives.
After this operation, 3,519 kB disk space will be freed.
Get:1 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 libpq-dev amd64 18.3-1.pgdg24.04+1 [167 kB]
Get:2 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 libpq5 amd64 18.3-1.pgdg24.04+1 [255 kB]
Get:3 https://packages.cloud.google.com/apt cloud-sdk/main all google-cloud-cli-app-engine-python all 564.0.0-0 [4,032 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libarchive13t64 amd64 3.7.2-2ubuntu0.6 [382 kB]
Get:5 https://packages.cloud.google.com/apt cloud-sdk/main amd64 google-cloud-cli-app-engine-go amd64 564.0.0-0 [3,007 kB]
Get:6 https://packages.cloud.google.com/apt cloud-sdk/main all google-cloud-cli-app-engine-java all 564.0.0-0 [160 MB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgdk-pixbuf2.0-common all 2.42.10+dfsg-3ubuntu3.3 [8,302 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgdk-pixbuf-2.0-0 amd64 2.42.10+dfsg-3ubuntu3.3 [147 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ruby3.2-dev amd64 3.2.3-1ubuntu0.24.04.7 [399 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libruby3.2 amd64 3.2.3-1ubuntu0.24.04.7 [5,341 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-libc-dev amd64 6.8.0-107.107 [2,096 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-xml amd64 8.3.6-0ubuntu0.24.04.8 [126 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.8 [13.5 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.8 [371 kB]
```

Step 2: Install Required Dependencies

```
g2022_anuprita_mhapankar@cloudshell:~$ sudo apt install -y jq curl git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3ubuntu0.24.04.1).
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
g2022_anuprita_mhapankar@cloudshell:~$
```

Then verify everything is present:

```
node
-v npm
-v
git --version
docker --version
docker ps
```

```
g2022_anuprita_mhapankar@cloudshell:~$ node -v
v24.14.1
g2022_anuprita_mhapankar@cloudshell:~$ npm -v
11.12.1
g2022_anuprita_mhapankar@cloudshell:~$ git --version
git version 2.43.0
g2022_anuprita_mhapankar@cloudshell:~$ docker --version
Docker version 29.4.0, build 9d7ad9f
g2022_anuprita_mhapankar@cloudshell:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 3: Fix Docker Compose

```
DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins
curl -SL
https://github.com/docker/compose/releases/download/v2.24.5/docker-compose
-linux-x86_64 \
  -o $DOCKER_CONFIG/cli-plugins/docker-compose
chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose
docker compose version
sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null ||
true
sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
#!/bin/bash
exec docker compose "$@"
EOF
sudo chmod +x /usr/local/bin/docker-compose
docker compose version
```

docker-compose version

```
g2022_anuprita_mhapankar@cloudshell:~$ DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins

curl -SL https://github.com/docker/compose/releases/download/v2.24.5/docker-compose-linux-x86_64 \
  -o $DOCKER_CONFIG/cli-plugins/docker-compose

chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose

docker compose version
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 58.5M  100 58.5M    0     0  79.4M      0 --:--:-- --:--:-- --:--:-- 79.4M
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$ sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null || true
g2022_anuprita_mhapankar@cloudshell:~$ sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
#!/bin/bash
exec docker compose "$@"
EOF
sudo chmod +x /usr/local/bin/docker-compose
g2022_anuprita_mhapankar@cloudshell:~$ docker compose version
docker-compose version
Docker Compose version v2.24.5
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 4: Clone Fabric Samples

cd $HOME

git clone https://github.com/hyperledger/fabric-samples

cd fabric-samples

```
g2022_anuprita_mhapankar@cloudshell:~$ cd $HOME
g2022_anuprita_mhapankar@cloudshell:~$ git clone https://github.com/hyperledger/fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15618, done.
remote: Counting objects: 100% (238/238), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 15618 (delta 161), reused 109 (delta 109), pack-reused 15380 (from 3)
Receiving objects: 100% (15618/15618), 24.07 MiB | 17.13 MiB/s, done.
Resolving deltas: 100% (8809/8809), done.
g2022_anuprita_mhapankar@cloudshell:~$ cd fabric-samples
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 5: Install Fabric Binaries and Docker Images

curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install- fabric.sh

chmod +x install-fabric.sh

./install-fabric.sh docker samples binary

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ chmod +x install-fabric.sh
./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.15 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz
===> Will unpack to: /home/g2022_anuprita_mhapankar/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100  120M  100  120M    0     0  22.5M      0  0:00:05  0:00:05 --:--:-- 24.4M
==> Done.
===> Downloading version 1.5.17 platform specific fabric-ca-client binary
===> Downloading:  https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz
===> Will unpack to: /home/g2022_anuprita_mhapankar/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 31.6M  100 31.6M    0     0  15.9M      0  0:00:01  0:00:01 --:--:-- 31.6M
==> Done.

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
===> Pulling fabric Images
====>  ghcr.io/hyperledger/fabric-peer:2.5.15
```

```
====>  ghcr.io/hyperledger/fabric-ca:1.5.17
1.5.17: Pulling from hyperledger/fabric-ca
18afb8c8d2a4: Pull complete
6f4ebca3e823: Pull complete
b9649d3f054a: Pull complete
0b966e606d9a: Pull complete
c36bacd85367: Download complete
Digest: sha256:453a0a727e82c4960b797e379adc231e853ee23ae7526db53afef77b5074117e
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17
ghcr.io/hyperledger/fabric-ca:1.5.17
===> List out hyperledger images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15    654bd4554bf9        250MB         80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17        453a0a727e82        381MB          123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15     86dded7eda92          1GB          255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15   9972c209be47        178MB         49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15      3e25adc31a75        230MB         66.2MB
hyperledger/fabric-baseos:2.5               654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:2.5.15            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:latest            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-ca:1.5                   453a0a727e82        381MB          123MB
hyperledger/fabric-ca:1.5.17                453a0a727e82        381MB          123MB
hyperledger/fabric-ca:latest                453a0a727e82        381MB          123MB
hyperledger/fabric-ccenv:2.5                86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:2.5.15             86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:latest             86dded7eda92          1GB          255MB
hyperledger/fabric-orderer:2.5              9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:2.5.15           9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:latest           9972c209be47        178MB         49.5MB
hyperledger/fabric-peer:2.5                 3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:latest              3e25adc31a75        230MB         66.2MB
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 6: Verify Docker images

```
docker images | grep fabric
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ docker images | grep fabric
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15    654bd4554bf9        250MB         80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17        453a0a727e82        381MB          123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15     86dded7eda92          1GB          255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15   9972c209be47        178MB         49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15      3e25adc31a75        230MB         66.2MB
hyperledger/fabric-baseos:2.5               654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:2.5.15            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:latest            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-ca:1.5                   453a0a727e82        381MB          123MB
hyperledger/fabric-ca:1.5.17                453a0a727e82        381MB          123MB
hyperledger/fabric-ca:latest                453a0a727e82        381MB          123MB
hyperledger/fabric-ccenv:2.5                86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:2.5.15             86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:latest             86dded7eda92          1GB          255MB
hyperledger/fabric-orderer:2.5              9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:2.5.15           9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:latest           9972c209be47        178MB         49.5MB
hyperledger/fabric-peer:2.5                 3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:latest              3e25adc31a75        230MB         66.2MB
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 7: Set Environment Variables

```
export PATH=$PATH:$HOME/fabric-samples/bin
export FABRIC_CFG_PATH=$HOME/fabric-samples/config
peer version
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export PATH=$PATH:$HOME/fabric-samples/bin
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ peer version
peer:
 Version: v2.5.15
 Commit SHA: 83c7930
 Go version: go1.26.0
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger

g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 8: Start the Network and Create Channel

```
cd $HOME/fabric-samples/test-network
./network.sh up createChannel
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export PATH=$PATH:$HOME/fabric-samples/bin
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ ./network.sh up createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'cryptogen'
Bringing up network
LOCAL_VERSION=v2.5.15
DOCKER_IMAGE_VERSION=v2.5.15
/home/g2022_anuprita_mhapankar/fabric-samples/test-network/../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] Running 3/7
 ⠿ Network fabric_test                      Created
 ⠿ Volume "compose_peer0.org1.example.com"  Created
 ⠿ Volume "compose_peer0.org2.example.com"  Created
 ⠿ Volume "compose_orderer.example.com"     Created
 ✔ Container orderer.example.com            Started
 ✔ Container peer0.org1.example.com         Started
 ✔ Container peer0.org2.example.com         Started
CONTAINER ID   IMAGE                               COMMAND             CREATED        STATUS                  PORTS
        NAMES
```

```
/home/g2022_anuprita_mhapankar/fabric-samples/test-network/../bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-04-09 14:56:55.392 UTC 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-04-09 14:56:55.402 UTC 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type: etcdraft
2026-04-09 14:56:55.402 UTC 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.EtcdRaft.Options unset, setti
n_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapshot_interval_size:16777216
2026-04-09 14:56:55.402 UTC 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/g2022_anuprita_mhapankar/fa
configtx.yaml
2026-04-09 14:56:55.408 UTC 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2026-04-09 14:56:55.408 UTC 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel genesis block
2026-04-09 14:56:55.410 UTC 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
        "name": "mychannel",
        "url": "/participation/v1/channels/mychannel",
        "consensusRelation": "consenter",
        "status": "active",
        "height": 1
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
```

```
Generating anchor peer update transaction for Org2 on channel mychannel
++ jq '.channel_group.groups.Application.groups.Org2MSP.values += {"AnchorPeers":{"mod_policy": "Admins","value":{"anchor_peers": [{"host": "peer0.org2.example.com","port": 90
51}]},"version": "0"}}' /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json
++ res=0
++ configtxlator proto_encode --input /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json --type common.Config --output /home/g2022
_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/original_config.pb
++ configtxlator proto_encode --input /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/Org2MSPmodified_config.json --type common.Config --output /h
ome/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/modified_config.pb
++ configtxlator compute_update --channel_id mychannel --original /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/original_config.pb --updated /ho
me/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/modified_config.pb --output /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifa
cts/config_update.pb
++ configtxlator proto_decode --input /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/config_update.pb --type common.ConfigUpdate --output /home/g
2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/config_update.json
++ jq .
+++ cat /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/config_update.json
++ echo '{"payload":{"header":{"channel_header":{"channel_id":"mychannel", "type":2}},"data":{"config_update":{' '"channel_id":' '"mychannel",' '"isolated_data":' '{},' '"read
_set":' '{' '"groups":' '{' '"Application":' '{' '"groups":' '{' '"Org2MSP":' '{' '"groups":' '{},' '"mod_policy":' '"",' '"policies":' '{' '"Admins":' '{' '"mod_policy":' '""
,' '"policy":' null, '"version":' '"0"' '},' '"Endorsement":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Readers":' '{' '"mod_policy":' '"",' '"poli
cy":' null, '"version":' '"0"' '},' '"Writers":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '}' '},' '"values":' '{' '"MSP":' '{' '"mod_policy":' '"",' '"v
alue":' null, '"version":' '"0"' '}' '},' '"version":' '"0"' '}' '},' '"mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '},' '"mod_policy":' '
"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '},' '"write_set":' '{' '"groups":' '{' '"Application":' '{' '"groups":' '{' '"Org2MSP":' '{' '"groups":' '{},' '
"mod_policy":' '"Admins",' '"policies":' '{' '"Admins":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Endorsement":' '{' '"mod_policy":' '"",' '"polic
y":' null, '"version":' '"0"' '},' '"Readers":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Writers":' '{' '"mod_policy":' '"",' '"policy":' null, '"
version":' '"0"' '}' '},' '"values":' '{' '"AnchorPeers":' '{' '"mod_policy":' '"Admins",' '"value":' '{' '"anchor_peers":' '[' '{' '"host":' '"peer0.org2.example.com",' '"por
t":' 9051 '}' '}' '},' '"version":' '"0"' '},' '"MSP":' '{' '"mod_policy":' '"",' '"value":' null, '"version":' '"0"' '}' '},' '"version":' '"1"' '}' '},' '"mod_policy":' '"",
' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '},' '"mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '}}}}'
++ configtxlator proto_encode --input /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/config_update_in_envelope.json --type common.Envelope --outp
ut /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/Org2MSPanchors.tx
2026-04-09 14:57:06.953 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-04-09 14:57:06.975 UTC 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 9: Deploy Chaincode

```
./network.sh deployCC \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccl javascript
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network

./network.sh deployCC \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccl javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --lang node --label basic_1.0
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
```

```
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
Using organization 1
Using organization 2
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/g2022_anuprita_mhapankar/fabric-samples/test-n
organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /ho
2_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 -
tCertFiles /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem --version 1
quence 1
+ res=0
2026-04-09 15:02:00.641 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)
alhost:7051
2026-04-09 15:02:00.646 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)
alhost:9051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 10: Set Peer Environment Variables

```
export FABRIC_CFG_PATH=$HOME/fabric-samples/config

export PATH=$PATH:$HOME/fabric-samples/bin

export CORE_PEER_TLS_ENABLED=true

export CORE_PEER_LOCALMSPID="Org1MSP"

export
CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizati
on
s/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca
.c rt
```

```
export
CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/
pe erOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 11: Initialize Ledger

```
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/exa
mp
le.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cer
t. pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.e
xa mple.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.e
xa mple.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function":"InitLedger","Args":[]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function":"InitLedger","Args":[]}'
2026-04-09 15:04:00.450 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 12: Query All Assets

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"doc
Type":"asset"},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"},{"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":
"Max","Size":10,"docType":"asset"},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"},{"AppraisedValue":800,"Color":"white","ID
":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

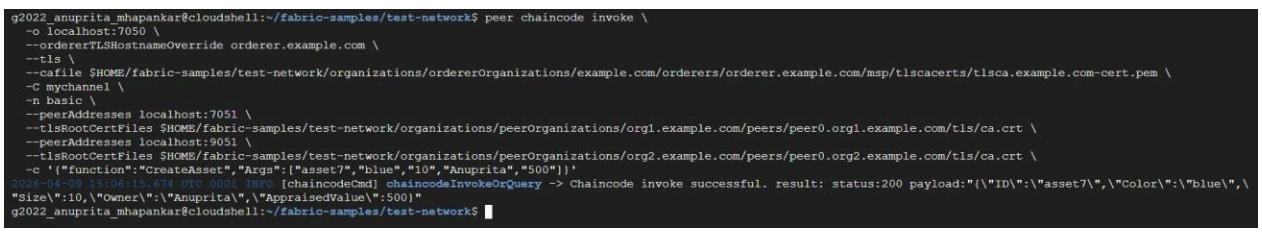
Step 13: Read a Specific Asset

```
peer chaincode query -C mychannel -n basic -c
'{"Args":["ReadAsset","asset5"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset5"]}'
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 14: Create a New Asset

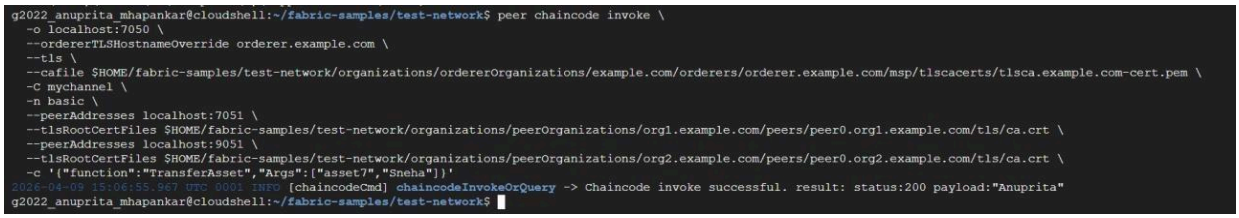
```
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/exa
mp
le.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cer
t. pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.e
xa mple.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.e
xa mple.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c
'{"function":"CreateAsset","Args":["asset7","blue","10","Anuprita","500"
]} '
```



Step 15: Transfer Asset

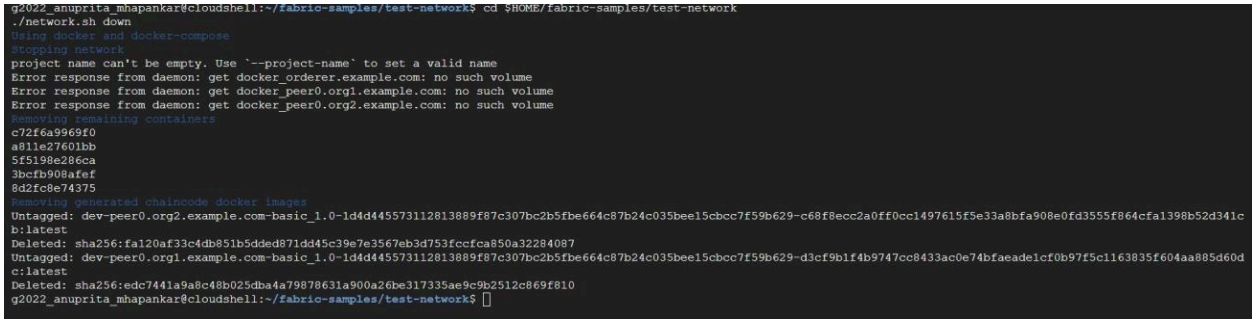
```
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/exa
mp
le.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cer
t. pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.e
xa mple.com/peers/peer0.org1.example.com/tls/ca.crt \
```

```
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.e
xa mple.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function":"TransferAsset","Args":["asset7","Sneha"]}'
```



Step 16: Stop the Network

```
cd $HOME/fabric-samples/test-network
./network.sh down
```



CONCLUSION:

The Hyperledger Fabric network was successfully implemented and chaincode was deployed. Asset operations such as creation, querying, and transfer were performed successfully. This experiment demonstrated secure and efficient asset management using a permissioned blockchain system.